SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1707**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

<u>Descriptive Written Test</u>

Code of Conduct:

I Geferlin Boffy R (192571080) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS

1. Smart Home Automation System

Introduction

A smart home automation system controls lighting, temperature, and security by using sensors and intelligent decision-making. It responds to real-time inputs such as motion detection, room temperature, and time of day while learning user preferences to improve comfort, energy efficiency, and security.

Types of Intelligent Agents

1. Simple Reflex Agent

Working

- Makes decisions based only on the current input.
- Uses condition-action rules.
- Does not remember previous events.

Application in Smart Home

- Turns ON lights when motion is detected.
- Turns OFF lights when no motion is detected.
- Activates an alarm when an intruder is detected.

Advantages

- Fast response.
- Easy to design.
- Suitable for simple tasks.

Disadvantages

- Cannot learn user preferences.
- Cannot handle complex situations.
- No memory of previous actions.

2. Model-Based Reflex Agent

Working

- Maintains an internal model of the environment.
- Uses current input and previous information to make decisions.

Application in Smart Home

- Remembers which rooms are occupied.
- Adjusts lighting based on occupancy.
- Controls temperature by considering previous room conditions.

Advantages

- Handles partially observable environments.
- Makes better decisions than a simple reflex agent.
- Stores information about the environment.

Disadvantages

- More complex than a simple reflex agent.
- Requires additional memory.

3. Goal-Based Agent

Working

- Selects actions to achieve a specific goal.
- Evaluates different possible actions before making a decision.

Application in Smart Home

- Maintains the desired room temperature.
- Ensures home security.
- Reduces electricity consumption while maintaining comfort.

Advantages

- Produces goal-oriented decisions.
- Can solve more complex problems.
- Considers future outcomes.

Disadvantages

- Slower than reflex agents.
- Requires planning before taking action.

4. Utility-Based Agent
Working
- Chooses the action with the highest utility (overall benefit).
- Considers multiple factors before making a decision.

Application in Smart Home
- Balances comfort and energy consumption.
- Selects the most energy-efficient temperature setting.
- Adjusts lighting according to user preferences and electricity usage.

Advantages
- Produces optimal decisions.
- Balances multiple objectives.
- Improves user satisfaction.

Disadvantages
- More computationally expensive.
- Utility calculation increases complexity.

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learning Ability	Smart Home Example
Simple Reflex	No	No	No	Turn lights ON when motion is detected
Model-Based	Yes	No	Limited	Remember occupied rooms
Goal-Based	Yes	Yes	Limited	Maintain desired temperature
Utility-Based	Yes	Yes	Yes	Balance comfort, security, and energy efficiency

Most Suitable Approach
A **Hybrid Approach** combining **Model-Based Agent** and **Utility-Based Agent** is the most effective.
Justification
- Maintains information about the home environment.
- Learns user preferences over time.
- Balances comfort and energy efficiency.
- Improves security by responding intelligently.
- Makes optimal decisions under different conditions.

Conclusion
A hybrid system using **Model-Based** and **Utility-Based Agents** provides the best performance because it remembers previous information, learns user preferences, optimizes energy usage, and ensures security and comfort.

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2. Robot Navigation in an Unknown Maze
Introduction
A robot is placed in an unknown maze and must find the exit without knowing the maze layout. It explores the environment and searches for the best path while avoiding dead ends.

Uniform Cost Search (UCS)
Working
- Expands the node with the lowest path cost.
- Uses a priority queue.
- Continues until the goal is reached.

Steps
1. Start from the initial position.
2. Insert the start node into the priority queue.
3. Expand the node with the lowest cost.
4. Add neighbouring nodes with updated costs.
5. Repeat until the exit is found.

Advantages
- Finds the optimal path.

- Considers different path costs.
- Works well when movement costs are unequal.

Disadvantages

- Uses large memory.
- Slower for large search spaces.
- Expands many unnecessary nodes.

Iterative Deepening Search (IDS)

Working

- Combines Depth First Search and Breadth First Search.
- Searches one depth level at a time.
- Increases the depth limit after each iteration.

Steps

1. Search with depth limit 0.
2. Increase depth to 1.
3. Continue increasing the limit.
4. Stop when the goal is found.

Advantages

- Requires less memory.
- Finds the shallowest solution.
- Suitable when search depth is unknown.

Disadvantages

- Repeats node expansion.
- Higher computation due to repeated searches.
- Less efficient than UCS when path costs differ.

Time and Space Complexity

Algorithm	Time Complexity	Space Complexity	Optimal
Uniform Cost Search	O(b^(1+C*/ε))	O(b^(1+C*/ε))	Yes
Iterative Deepening Search	O(b^d)	O(b × d)	Yes (Equal Cost)

Where:

- **b** = Branching factor
- **d** = Depth of the solution
- **C*** = Cost of the optimal solution
- **ε** = Minimum step cost

Relation to Intelligent Agents

Simple Reflex Agent

- Acts only on the current perception.
- Cannot remember explored paths.
- Not suitable for unknown mazes.

Model-Based Agent

- Stores visited locations.
- Builds an internal map of the maze.
- Avoids revisiting the same paths.

Goal-Based Agent

- Searches specifically for the exit.
- Plans movements to reach the goal efficiently.

Utility-Based Agent

- Selects paths that minimize cost and risk.

- Chooses the most beneficial route among alternatives.

Effect of Agent Design on Decision-Making

- **Simple Reflex Agent:** Makes quick decisions but may get trapped in loops.
- **Model-Based Agent:** Uses memory to improve navigation.
- **Goal-Based Agent:** Plans actions to reach the exit efficiently.
- **Utility-Based Agent:** Selects the safest and least costly path.

Most Effective Combination

Agent Type: Goal-Based Agent with Model-Based features
Search Strategy: Uniform Cost Search (UCS)

Justification

- Stores information about explored paths.
- Plans movements toward the exit.
- Finds the minimum-cost path.
- Avoids revisiting explored locations.
- Produces optimal and efficient navigation.

Conclusion

For an unknown maze, the combination of a **Goal-Based Agent with Model-Based capabilities** and **Uniform Cost Search (UCS)** is the most effective solution because it remembers explored areas, searches intelligently for the exit, minimizes path cost, and makes better decisions in an unknown environment.